MILITARY SPECIFICATION

INDICATOR SET, VERTICAL SCALE

This specification is approved for use by the
Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope. The equipment covered by this specification is a tape type vertical scale indicator set for engine performance indication.

1.2 Classification. The indicator set covered by this specification shall consist of one each of the following dual indicators:

Type Designation	Indicator
EHU-41/A	Indicator, Exhaust Gas Temperature (EGT)
ERU-20/A	Indicator, Electrical Tachometer (RPM)
EFU-45/A	Indicator, Rate of Fuel Flow (FF)

1.3 Associated Equipment. The indicator set shall operate from inputs from the associated equipment listed in 3.3.19.

2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation of bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-B-5087	Bonding, Electrical and Lighting Protection, for Aerospace Systems
MIL-W-5088	Wiring, Aircraft, Selection and Installation of

COMMENTS OR RECOMMENDATIONS WHICH MAY BE OF USE IN IMPROVING THIS DOCUMENT MAY BE SENT TO THE PREPARING ACTIVITY WHEN A SELF-ADDRESSED STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL (DD FORM 1426) APPEARS AT THE END OF THIS DOCUMENT, OR BY LETTER ADDRESSED TO THE COGNIZANT ACTIVITY.

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MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-W-5846	Wire, Electrical, Chromel and Alumel, Thermocouple
MIL-I-7071	Indicator, Temperature, Thermocouple, Hermetically Sealed, General Specification for
MIL-C-14806	Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges
MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment, Including Ground Support Equipment
MIL-T-22030	Transmitter, Rate of Flow, Fuel, TRU-26/A (300 to 10,000 PPH)
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-G-26611	Generator, Tachometer, GEU-7/A, Miniature
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-S-38130	System Safety Engineering of Systems and Associated Subsystems and Equipment; General Requirements for
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge, and Cover

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirement for
AR-34	Failure Classification of Reliability Testing, General Requirement for

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STANDARDS

Federal

Federal Standard Number 595	Colors

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-143	Standards and Specification, Order and Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristic, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electronic Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests; Expotential Distribution
MIL-STD-785	Requirements for Reliability Program
MIL-STD-794	Parts and Equipment, Procedures for Packing and Packaging of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form for

2.1.1 Availability of Documents. When requesting specifications, standards, drawings, and publications, refer to both title and number. Copies of specifications, standards, and drawings required by suppliers in conjunction with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 First Article Testing. This specification makes provision for first article testing (4.3).

3.2 Parts, Materials, and Processes. In the selection of parts, materials, and processes, the fulfillment of major design objectives shall be of prime consideration. Parts, materials, and processes shall conform to the requirements of MIL-E-5400, the specifications and standards listed in MIL-STD-143, and the requirements listed herein. Electron tubes shall not be used. When suitable parts and materials have not been established in a military standard, such parts and materials shall be used which are of the best commercial quality, of the lightest practicable weight, entirely suited for the purpose and readily available. However, only nonflammable materials shall be used. Commercial utility parts such as screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. Such parts can be replaced by the military standard parts without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractors drawings.

(Military standard parts shall be identified on the drawings by their part numbers.)

3.2.1 Nonstandard Parts and Materials Approval. Approval for the use of nonstandard parts and materials of commercial quality (see 3.2), including transistors and diodes other than microelectronic devices, shall be obtained as outlined in MIL-E-5400.

3.2.2 Microelectronic Devices and Modular Assemblies. Microelectronic devices shall be selected and approved in accordance with AR-5. When used, microelectronic modular assemblies shall be described by a contractor specification and must be approved by the procuring activity. Conformal coatings, encapsulants, embedments, or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.2.3 Part Testing. No part shall be used in the equipment without evidence of the part having been tested. The test results may be available from a data exchange program, from previous contracts, or if not so available, the parts must be tested under this requirement. Unless required by the contract, the submission of the test information on parts is not required, but the manufacturer shall be able to produce test results when requested by the procuring activity.

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3.2.4 Part Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.2.5 Magnesium and Magnesium Alloys - Magnesium and magnesium alloys shall not be used unless significant weight savings, performance, or other advantage is obtained; in which case, each application shall be subject to the approval of the procuring activity. Quantitative substantiating data shall be submitted with the application.

3.2.6 Nonmagnetic Materials - Nonmagnetic materials shall be used for all parts of the indicator set except where magnetic materials are essential.

3.2.7 Metals. Metals shall be of the corrosion-resistant type or suitably protected to resist corrosion during normal service life.

3.2.8 Dissimilar Metals. Dissimilar metals as defined by MIL-STD-889 shall not be used in intimate contact with each other unless suitably protected against electrolytic corrosion by means of protective coating or sealant.

3.2.9 Fungi-Resistant Materials. Materials which do not support fungal growth shall be used to the greatest extent practicable. In cases where fungal nutrient material must be used, such material shall be treated with a suitable fungicidal agent or be protected by hermetic sealing.

3.2.10 Corrosive Fumes. The components used in the construction of the indicator set shall be selected such that no volatiles of potentially toxic or corrosive nature will be liberated during the tests specified in 4.7 or due to failure of a component.

3.2.11 Filling Medium - The filling medium shall be a mixture of 90 $\pm$ 2 percent nitrogen and the remainder helium. The filling medium shall be at least 98 percent pure, be free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dewpoint -65° Celsius (C)) at the filling pressure. The absolute pressure of the filling medium in the hermetically sealed enclosure shall be approximately 1 atmosphere.

3.2.12 Protective Treatment. Exposed materials used in the construction of the indicator set which are subject to corrosion in salt air or other atmospheric conditions likely to be encountered during service usage shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall not be used.

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3.2.13 Jewel Bearings. When jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.3 Design, Construction, and Workmanship.

3.3.1 General. The equipment shall contain all of the instrumentation required to process and display the signal data received from the engine mounted sensors. The detailed mechanical and electrical design of the set indicators shall be accomplished by the manufacturer in accordance with MIL-E-5400 for class 2 equipment, subject to the requirements of this specification. The requirements of this specification are detailed only to the extent considered necessary to obtain the essential mechanical and electrical characteristics, performance, and reliability of the indicator set. The design, layout, and assembly of the indicators and their component parts shall be such as to facilitate quantity production, failure analysis and repair, and to result in minimum power consumption and weight.

3.3.2 Modularization. The electronic portions of the equipment shall be functionally modularized in accordance with AR-10 and as outlined in MIL-E-5400.

3.3.2.1 Modification. When a manufacturer has previously produced equipment to this specification, the design shall not be changed without the written approval of the procuring activity.

3.3.3 Indicator Case. The indicator case shall be as specified in the detail requirements and designed to permit mounting in a mounting rack assembly in accordance with Figure 8. The cases shall be hermetically sealed and shall have a smooth external surface. The cases shall be so designed that the internal mechanisms may be removed, replaced, and the cases resealed without the use of special tools and fixtures. If tear strips are provided for this purpose they shall not have end pieces protruding outside the case dimensions. Filling tubes, if used, shall not be left exposed to possible damage from impact or handling. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the indicator may be subjected. Elastromeric or resinous materials may be used for cover glass seals provided the specified leak rate requirements are met under all tests and environmental conditions specified herein. The cases shall be finished in a durable dull black conforming to Federal Standard Number 595 (FED-STD-595), color number 37038.

3.3.4 Indicator Mechanisms. The indicator mechanisms shall be sufficiently rugged that small distortions of the case will not affect their accuracy of operation.

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3.3.5 Total Weight. The total weight of an indicator set, including mounting rack, but excluding interconnection cabling, shall be a minimum consistant with good design and shall not exceed 9.0 pounds.

3.3.6 Integral Lighting. Integral lighting of the indicators shall be in accordance with MIL-L-25467.

3.3.6.1 Lamp Location. The lighting system shall be so designed that when lamp replacement is necessary the lamps may be accurately and securely positioned. No provision shall be made for lamp replacement in the field, but the lamps shall be so installed that easy replacement be accomplished at a repair depot.

3.3.7 Cover Glass. The cover glass shall be in accordance with MIL-G-81704. The thickness of the glass shall be the minimum practicable and entirely suitable for the purpose. Any allowable flaws shall not interfere with the readability of the display.

3.3.7.1 Reflection-Reducing Coating. All reflecting surfaces of transparent materials incorporated in the display and the cover glass shall be provided with a reflection reducing coating that conforms with the requirements of MIL-C-14806.

3.3.8 Display Visibility. All portions of the scale and tapes shall be visible from any point within the frustrum of a wedge whose sides make an angle of 30 degrees with a perpendicular to the display surface and whose small parallel surface is the aperature of the case.

3.3.8.1 Scale to Cover Glass Distance. The maximum distance between the inner surface of the cover glass and the indicator scale shall be 0.062 inch.

3.3.8.2 Anti-Parallax. The indicating tapes shall be located in the same plane as the calibration marks within a maximum dimension of 0.005 inch in order to minimize parallex errors.

3.3.9 Display Presentation. The scales of the indicators shall be as specified in the detail requirements, with letters and numerals in accordance with MS33558. The indicator parameters shall be indicated by changing the height of the white tape against a fixed, calibrated center scale. The top of the white tape shall be aligned with respect to the graduated center scale in accordance with the value of the input signal for that channel. Scale markings and the white tape shall be lusterless white, color number 37875 of FED-STD-595. The display and tape backgrounds shall be lusterless black, color number 37038 of FED-STD-595.

3.3.10 Power Off Warning. Each tape of the indicator shall incorporate a power-off warning. Each channel shall be so mechanized that upon interruption of the 26 volt (v), 400 hertz (Hz) power, the tape will drive to an off-scale position in the direction of decreasing indication and display

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the word "OFF" within 5 seconds. The word "OFF" shall be in gloss black letters, color number 17038 of FED-STD-595 on a lusterless white background, color number 37875 of FED-STD-595.

3.3.11 Tape Drive Stops. The display drive mechanism shall incorporate lower and upper stops that will limit tape travel from the power-off warning position to a position between 1/32 and 3/32 inch (0.8 and 2.4 mm) beyond the maximum range of the parameter being displayed. Repeated striking of the limit stops shall not result in damage to the mechanism.

3.3.12 Signal Failure. Interruption of input signal to a channel by either open or short circuiting shall cause the tape to drive to zero indication. The time shall not exceed 5 seconds to attain the zero from full scale indication.

3.3.13 Reliability. The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder for a supplier who has previously produced the indicator set, the program previously used may be continued unless otherwise required by the contract or order.

3.3.13.1 Operational Stability. The indicators shall meet the reliability and accuracy requirements as specified herein without necessity for readjustment. No external means of adjustment shall be provided. The indicator mechanisms shall be properly stabilized to prevent hunting and damped to prevent more than one overshoot of the tape. Stability shall be within specified limits for both continuous and intermittent operation for a period of at least 3500 hours.

3.3.13.2 Longevity. The indicators shall have a total operating life of at least 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.13.3 Reliability in Mean-Time-Between Failures (MTBF). The indicators shall be designed to provide the following specified MTBF (θ_o as defined in MIL-STD-781).

Indicator	Specified Hrs. MTBF
Indicator, Electrical Tachometer (RPM), Type ERU-20/A	7000
Indicator, Exhaust Gas Temperature (EGT), Type EHU-41/A	7000
Indicator, Rate of Fuel Flow (FF), Type EFU-45/A	7000

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3.3.14 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.15 Interference Control. The emission of interference by the indicator set and susceptibility of the indicators to interference shall be controlled within the limits of MIL-STD-461, Table II, for Class ID equipment for the frequency range 0.15 to 500 MHz.

3.3.16 Bonding. The equipment shall be bonded in accordance with MIL-B-5087, Class H (shock hazard) bonding.

3.3.17 Cooling. The equipment shall be cooled by free convection and radiation to the ambient environment.

3.3.18 Safety. The equipment shall be designed to preclude the incorporation of features which would result in critical or catastropic hazards as classified in MIL-S-38130, hazard conditions classification.

3.3.19 Signal Inputs. The indicator set shall be designed to operate from the following input signals:

RPM — GEU-7/A tachometer generator output in accordance with MIL-G-26611, 4200 revolutions per minute (rpm) = 100% RPM

EGT — Direct current (DC) millivoltage from a Type K (Chromel-Alumel) thermocouple harness per MIL-I-7071

FF — TRU-26/A rate of fuel flow transmitter synchro output in accordance with MIL-T-22030

3.3.20 Cabling and Connections. Cabling and connections shall be as follows:

3.3.20.1 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.3.20.2 Interconnecting Cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded. External cables and that portion of the connectors attached to the cables shall not be part of the equipment.

3.3.20.3 Indicator Connections. Indicator connections shall be as shown in the system wiring diagram, Figure 4.

3.3.21 Nomenclature, Nameplates, and Identification Marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307, except that the nameplate shall be located on the rear of each indicator. Decalcomanias shall not be used.

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3.3.22 Input Power Requirements. The indicator set shall give specified performance from power sources with characteristics as defined in MIL-STD-704, having limits as specified herein. The power required shall not exceed specified limits.

a. Alternating current (ac) power (single phase) 26v, 400 Hz, Category B, 42 volt-amperes (va).

b. Ac power (Lighting) 5v, 400 Hz, 9.0 va.

The power factor of the ac power circuits shall be within 0.85 lagging to 0.95 leading.

3.3.22.1 Overload Protection. Overload protection for the equipment shall be provided integrally. All parts and circuits of the equipment which are likely to carry an overload due to failures or poor adjustment shall be proportioned to withstand an overload without permanent damage to the equipment or shall have suitable protective devices. The use of fuses is forbidden, and other protective devices shall be held to a minimum.

3.3.22.2 Undervoltage Protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3.23 Workmanship. The indicators, including all parts and assemblies shall be constructed and finished in a thoroughly workmanlike manner and shall conform to the requirements of MIL-STD-454.

3.4 Detail Requirements.

3.4.1 Exhaust Gas Temperature (EGT) Indicator. The EGT indicator shall conform to the following detail requirements.

3.4.1.1 Function. The EGT indicator shall include all of the circuitry necessary to convert signals from Type K (Chromel-Alumel) theromcouples located in the exhaust gases of the left and right turbojet engines to a dual temperature indication in degrees Celsius (C).

3.4.1.2 Form Factor. The EGT indicator shall conform to Figure 1.

3.4.1.3 Weight. The weight of the EGT idicator shall not exceed 2.7 pounds.

3.4.1.4 Display. The display shall conform to Figure 5.

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3.4.1.5 Electrical Connectors. The electrical connectors shall be the types specified below, shall be wired to provide specified operation when connected to power and signal inputs as shown on Figure 4, and shall be hermetically sealed. Pins A and C of the Cannon connector shall be made of alumel and pins B and D of chromel.

Reference Designation	Receptacle Type	Function
J1	Amphenol 67-04H14-9P(105)	Power Inputs
J2	Cannon GS02-14S-2P-003(F9)(F109)	Type K Thermocouple Inputs

3.4.1.6 Thermocouple Cold Junction Compensation. The indicator shall incorporate temperature compensation for the thermocouple cold junction within the indicator. Temperature control at the cold junction shall not be used. The compensation shall be such that the circuit is independent of the temperature changes at the cold junction. Cold junction reference temperature shall be either 24°C or 0°C. Reference junction temperature shall be included on the nameplate.

3.4.1.7 Reference Voltage. The reference voltage, if required for the self-balancing potentiometer, shall be derived from the power supply in the indicator. Standard cells shall not be used.

3.4.1.8 Performance. The EGT indicator shall conform to the following performance requirements in addition to those listed in 3.6.

3.4.1.8.1 Operation. The EGT indicator shall translate a millivoltage input signal from Type K thermocouples which are proportional to the temperature difference between the hot and cold temperature junctions and actuate the tape drives. The tapes shall be positioned to indicate the hot junction temperatures in degrees Celsius.

3.4.1.8.2 Stabilization. The indicator channels shall be stabilized with a damping ratio between 0.8 and 1.0 to prevent hunting and overshoot.

3.4.1.8.3 Scale Error. The maximum scale error allowed for the test points in the expanded range shall apply at any indication between 400°C and 700°C.

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TABLE I. EGT Indicator Scale Error Tolerances.

Sig. Input mv dc (0°C Ref. Jun.)	Test Points °C	Max. Scale Error in ± °C 1/			
		25°C	71°C	-54°C	Reliability (Qual. Phase Only)
0.00	0	15	20	25	20
8.13	200	15	20	25	20
16.82	410	5	7.5	10	10
20.64	500	5	7.5	10	10
24.90	600	5	7.5	10	10
28.71	690	5	7.5	10	10
37.36	900	15	20	25	20

NOTE:
1/ Scale Error tolerance shall be interpolated by the proportion 1°C = 0.043 millivolts (mv) for each test point in accordance with MIL-I-7071.

3.4.1.8.4 Position Error. Position error shall not exceed 2°C at any indication between 400°C and 700°C.

3.4.1.8.5 Response Time. Stabilized change in indication from 0°C to 900°C as a result of an instantaneous signal input change shall not exceed 5 seconds.

3.4.1.8.6 Warmup Period. Warmup and tape travel from OFF to 900°C shall not exceed 10 seconds as a result of an instantaneous application of power and 900°C signal inputs.

3.4.1.8.7 Vibration Error. The change in indication plus tape oscillation shall not exceed 8°C throughout the expanded scale range as a result of the vibration specified in 4.7.22.

3.4.1.8.8 Voltage and Frequency Variation Error. Steady state frequency variations ranging from 380 Hz to 420 Hz and voltage variations from 23.4v to 28.6v shall cause a change in indication no greater than 3°C.

3.4.1.8.9 Thermocouple Circuit Resistance Effect. Indication change shall not exceed 2°C at any indication between 400°C and 700°C when lead resistance changes from 5 ohms to 200 ohms.

3.4.1.8.10 Grounding of Thermocouple Extension Wires. Indicator steady state change shall not exceed 2°C at any indication between 400°C and 700°C when either the chromel or alumel conductor is grounded separately at any point on the ground side (low potential side) of the power supply.

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3.4.1.8.11 Stray Signal and Beat Frequency Toleration. The indicator shall withstand the effects of beat frequencies in the power supply and stray ac signals in the thermocouple circuit as specified in 4.7.19 with degradation of indication no greater than that listed therein.

3.4.2 Electrical Tachometer (RPM) Indicator. The RPM indicator shall conform to the following detail requirements.

3.4.2.1 Function. The RPM indicator shall convert electrical signals from one phase of the tachometer generators, driven by the accessory gear boxes of the left and right turbojet engines, to dual indication in percent RPM, where 100 percent RPM equals 4200 rpm tachometer generator shaft speed.

3.4.2.2 Form Factor. The RPM indicator shall conform to Figure 2.

3.4.2.3 Weight. The weight of the RPM indicator shall not exceed 2.7 pounds.

3.4.2.4 Display. The display shall conform to Figure 6.

3.4.2.5 Electrical Connector. The electrical connector shall be as specified below, shall be wired to provide specified operation when connected to power and signal inputs as shown in Figure 4, and shall be hermetically sealed.

Reference Designation	Receptacle Type	Function
J1	Amphenol 67-04H14-12P(105)	Power and Signal Inputs

3.4.2.6 Performance. The RPM indicator shall conform to the following performance requirements in addition to those listed in 3.6.

3.4.2.6.1 Operation. The RPM indicator shall translate the frequency of ac voltage inputs from one phase of tachometer generators in accordance with MIL-G-26611, which are proportional to the rotational speed of the high pressure compressors of the engines, and actuate the tape drives. The tapes shall be positioned to indicate these rotational speeds in percent RPM.

3.4.2.6.2 Stabilization. The indicator channels shall be stabilized with a damping ratio between 0.8 and 1.0 to prevent hunting, overshoot, and oscillation. The indication shall not be affected by a noise voltage of 2.0 root mean square (rms), 400 Hz on the signal input.

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3.4.2.6.3 Minimum Starting. The indicator shall begin functioning and provide stable indication with input signals no greater than 10 percent RPM from the tachometer generators.

3.4.2.6.4 Scale Error. The maximum scale error allowed for the test points in the expanded range shall apply for any indication between 70 percent and 110 percent RPM and that allowed for the test points in the contracted range shall apply for any indication between 10 percent and 70 percent RPM.

TABLE II. RPM Indicator Scale Error Tolerances.

Gen. Shaft rpm	Test Points % RPM	Max. Scale Error ± % RPM			
		25°C	71°C	-54°C	Reliability (Qual. Phase Only)
630	15	1.5	2.5	3.0	4.0
840	20	1.5	2.5	3.0	4.0
1680	40	1.5	2.5	3.0	4.0
2982	71	0.4	0.7	1.0	1.5
3360	80	0.4	0.7	1.0	1.5
3780	90	0.4	0.7	1.0	1.5
4200	100	0.4	0.7	1.0	1.5
4620	110	0.4	0.7	1.0	1.5

3.4.2.6.5 Position Error. Position error shall not exceed 0.2 percent RPM at any indication between 70 percent and 110 percent RPM.

3.4.2.6.6 Warmup Period. Warmup and tape travel from OFF to 110 percent RPM shall not exceed 10 seconds as a result of an instantaneous application of power and 110 percent signal inputs.

3.4.2.6.7 Response Time. Stabilized change in indication from 0 percent to 110 percent RPM shall not exceed 5 seconds as a result of an instantaneous input change.

3.4.2.6.8 Vibration Error. The change in indication plus tape oscillation shall not exceed 1.0 percent RPM throughout the expanded scale range as a result of vibration as specified in 4.7.22.

3.4.2.6.9 Voltage and Frequency Variation Error. Steady state frequency variations ranging from 380 Hz to 420 Hz and voltage variations from 23.4v to 28.6v shall cause a change in indication no greater than 0.4 percent RPM throughout the expanded scale range.

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3.4.3 Rate of Fuel Flow (FF) Indicator. The FF indicator shall conform to the following detail requirements.

3.4.3.1 Function. The FF indicator shall convert electrical signals from the synchro output of fuel flow transmitters in accordance with MIL-T-22030, located in the left and right turbojet engine fuel lines, to a dual indication in pounds per hour (pph).

3.4.3.2 Form Factor. The FF indicator shall conform to Figure 3.

3.4.3.3 Weight. The weight of the FF indicator shall not exceed 2.7 pounds.

3.4.3.4 Display. The display shall conform with Figure 7.

3.4.3.5 Electrical Connector. The electrical connector shall be as specified below, shall be wired to provide specified operation when connected to power and signal inputs as shown in Figure 4, and shall be hermetically sealed.

Reference Designator	Reference Type	Function
J1	Amphenol 67-04H20-37P(105)	Power and Signal Inputs

3.4.3.5 Performance. The FF indicator shall conform to the following performance requirements in addition to those listed in 3.6.

3.4.3.6.1 Operation. The FF indicator shall translate the synchro outputs from flowmeter transmitters in accordance with MIL-T-22030, which are functions of the fuel flow rates, and actuate the tape drives. The tapes shall be positioned to indicate these flow rates in pph.

3.4.36.2 Stabilization. The indicator channels shall be stabilized with a damping ratio between 0.8 and 1.0 to prevent hunting and overshoot.

3.4.3.6.3 Scale Error. The maximum scale error for the test points in the expanded range shall apply for any indication between 300 and 4900 pph.

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TABLE III. FF Indicator Scale Error Tolerances.

Fuel Flowmeter Synchro Transmitter Simulator Degrees	Fuel 1/ Flow Indicator pph	Max. Scale Error ± pph			
		25°C	71°C	-54°C	(Qual. Phase Only)
101.65	500	20	30	40	26
116.00	1000	20	30	40	26
144.70	2000	20	30	40	26
173.40	3000	20	30	40	26
179.14	3200	20	30	40	26
202.10	4000	20	30	40	26
227.93	4900	20	30	40	26
257.40	9000	100	150	200	135

NOTE:
1/ The Fuel Flow Indicator Synchro shall be wired to conform to the aircraft wiring of the Fuel Flow Transmitter Synchro as shown in Figure 4. The electrical zero (EZ) (low null at XY) of the Fuel Flow Transmitter Synchro and Fuel Flow Indicator Synchro corresponds to a fuel flow rate of 3230 pph. The direction of rotation at EZ is such that as the fuel flow rate increases, the output voltage XZ increases and the output voltage YZ decreases.

3.4.3.6.4 Position Error. Position error shall not exceed 10 pph at any indication between 300 and 4900 pph.

3.4.3.6.5 Warmup Period. Warmup and tape travel from OFF to 10,000 pph shall not exceed 10 seconds as a result of an instantaneous application of power and 10,000 pph signal inputs.

3.4.3.6.6 Response Time. Stabilized change in indication from 300 pph to 10,000 pph shall not exceed 5 seconds as a result of an instantaneous input change.

3.4.3.6.7 Vibration Error. The change in indication plus tape ocillation shall not exceed 20 pph throughout the expanded scale range as a result of vibration as specified in 4.7.22.

3.4.3.6.8 Voltage and Frequency Variation Error. Steady state frequency variations ranging from 380 Hz to 420 Hz and voltage variations from 23.4v to 28.6v shall cause a change no greater than 20 pph throughout the expanded scale range.

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3.5 Service Conditions. The indicator set shall be so designed and constructed that no part or assembly shall become loose and no degradation be caused in performance in excess of that specified herein during operation or after storage in the environmental conditions, or any natural combination of these conditions specified as follows.

3.5.1 Temperature.

3.5.1.1 Operating. The indicator set shall continuously operate as specified herein over the range of -54°C to +71°C (-65°F to +160°F), the temperature remaining constant for long periods or varying at rates to 1°C per second.

3.5.1.2 Nonoperating. In a nonoperating condition the indicator set shall be capable of withstanding long periods of exposure to the extremes of -65°C to +95°C (-85°F to +203°F) without subsequent degradation of performance.

3.5.2 Altitude. The indicator set shall continuously operate as specified over the range of -1000 feet to 70,000 feet.

3.5.3 Vibration. The indicator set shall withstand vibration levels as specified in 4.7.22 without:

a. Deviation of indication greater than that specified in 3.4.

b. Scale error after vibration exceeding the 25°C tolerances.

c. Looseness or damage to any part of the instruments.

3.5.4 Shock.

3.5.4.1 Basic Design. The indicator set shall not suffer damage or subsequently fail to provide the performance specified herein when subjected to 18 impact shocks of 15 times the acceleration due to gravity (g), consisting of 3 shocks in opposite directions along each of 3 mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The g value shall be within ± 15 percent of the nominal 15g and the waveform shall be half sine such that the maximum g occurs at approximately 5 1/2 milliseconds.

3.5.4.2 Crash Safety. The indicator set shall remain in place and not create a hazard when subjected to 12 impact shocks of 30g, consisting of 2 shocks in opposite directions along each of 3 mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 milliseconds.

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The g value shall be within ± 15 percent of the nominal 30g and the waveform shall be half sine such that a maximum g occurs at approximately 5 1/2 milliseconds. The indicators are not required to operate following these shocks and case distortion is allowed.

3.5.5 Salt Atmosphere. The indicator set shall withstand, in an operating and nonoperating condition, extended exposure to salt-sea atmosphere.

3.6 Performance. The indicator set shall conform to the following requirements when subjected to the tests specified in Section 4.

3.6.1 Power Consumption. The 26v, 400 Hz steady-state mechanism power consumption of each type indicator shall not exceed 9.5 va and the maximum dynamic power consumption shall not exceed 14 va. The 5v, 400 Hz lighting circuit power consumption shall not exceed 3 va.

3.6.2 Magnetic Effect. Compass deflection shall not exceed 3 degrees when tested as specified in 4.7.12.

3.6.3 Fogging. No evidence of water or oil fog shall be noted on the inner surface of the cover glass when tested as specified in 4.7.11.

3.6.4 Temperature Shock. The indicators shall withstand the temperature shock test specified in 4.7.26 without degradation of performance or deterioration of the case, hermetic seal, coatings, or assemblies.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as specified in the contract or order, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection. First Article Inspection shall be conducted on a sample consisting of the following number of indicators.

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a. Six indicators, consisting of an equal number of each type indicator on the contract, shall be subjected to the Reliability Qualification Phase Test by the supplier's test facility.

b. In addition, one of each type of indicator or a minimum of two indicators, as applicable, shall be subjected to the Individual Tests only by the supplier. These indicators, along the the Individual Test data shall then be forwarded at the supplier's expense to a Government laboratory designated by the procuring activity. The Government laboratory shall conduct all of the tests listed in 4.7 on these indicators.

4.3.1 First Article Test Data. The contractor shall submit a report in accordance with MIL-STD-781 of the Reliability Qualification Phase Test to the procuring activity. All data collected in conducting this test shall be made available to the procuring activity for review at Government request.

4.3.2 First Article Approval. Approval of the First Article sample shall be by the procuring activity upon satisfactory completion of all tests. No production indicators shall be delivered prior to approval of the First Article sample. Prefabrication of production indicators prior to approval of the First Article sample is at the supplier's own risk. The First Article sample shall not be delivered as production indicators on the contract.

4.3.3 Production Indicators. Indicators supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equivalent to the approved First Article sample indicators. Each indicator shall be capable of passing the same tests as imposed on the First Article sample indicators of that type. Evidence of noncompliance with the above shall be cause for rejection of all indicators on the contract.

4.4 Quality Conformance Inspection. The supplier shall furnish all samples and shall be responsible for accomplishing all the inspections except Sampling Plan "B" Tests, which will be conducted at a Government laboratory designated by the procuring activity. Quality conformance inspections, except Sampling Plan "B" Tests, shall be under the supervision of the Government Quality Assurance Representative. The supplier shall furnish test reports showing the detailed quantitative results of all tests required of the supplier. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. The quality conformance inspection shall consist of the following tests:

a. Reliability Qualification Phase Test

b. Reliability Production Acceptance Phase Test

c. Individual Tests

d. Sampling Plan "A" Tests

e. Sampling Plan "B" Tests

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extending more than 180 degrees around the circumference, or by an

In addition to the above tests, the quality conformance inspection shall include examination of the packaging, packing, and marking in preparation for delivery in accordance with the requirements of Section 5.

4.4.1 Acceptance for Delivery. Each indicator of the indicator set accepted for delivery on the contract or order shall have conformed to the Reliability Production Acceptance Phase Test including the Individual Tests requirements at the conclusion of that test.

4.4.2 Reliability Assurance Tests. The reliability assurance tests shall consist of the Reliability Qualification Phase Test and the Reliability Production Acceptance Phase Test. These tests shall be conducted in accordance with MIL-STD-781. Classification of failures shall be in accordance with MIL-STD-781 and AR-34.

4.4.2.1 Reliability Qualification Phase Test. Six indicators, consisting of an equal number of each type indicator on the contract, shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification (Demonstration) Phase of Product Reliability Tests." Test Plan IV and Test Level E shall be used. The six indicators shall be tested, without regard to type, to demonstrate a specified MTBF of 7000 hours.

4.4.2.2 Reliability Production Acceptance Phase Test. Each production indicator offered for delivery on the contract shall be subjected to this test. This test shall consist of a burn-in period, reliability test phase, and the individual tests at the conclusion of the reliability test phase.

4.4.2.2.1 Burn-In Period. A burn-in period shall be performed on each indicator before subjecting the indicator to the reliability test phase. The indicator shall be subjected to the same test level and duty cycle as the reliability test phase for a period of 40 hours. Only equipment "on-time" shall be counted toward the 40 hour requirement. When a failure occurs, the indicator shall be repaired, the failure and repair action documented and the test continued.

4.4.2.2.2 Reliability Test Phase. The indicators shall be tested as outlined in MIL-STD-781 under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests." Test Plan XXIX and Test Level E shall be used.

4.4.2.2.2.1 Procedure for the Reliability Test Phase. Each indicator shall be tested for 100 hours equipment "on-time." Time accrued during the burn-in period shall not be considered as part of the 100 hours. To determine if the MTBF is being met at any time during the contract, the operation test hours ("on-time") and the failures (not counting burn-in time or failures) shall be totaled and the results compared to the reject

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line of Test Plan XXIX of MIL-STD-781. (Extend the reject line as necessary to accomodate the data.) The specified MTBF for indicators of 3.3.13.3 shall apply. These totals shall be accumulative so that the experience from the beginning of the contract is always included. At the end of each month, the results shall be sent to the procuring activity. At any time the current total of test hours and test failures plotted on Test Plan XXIX curves show a reject situation, and test failures plotted on Test Plan XXIX show a reject situation, the procuring activity shall immediately be notified. The procuring activity reserves the right to stop the acceptance of indicators at any time a reject situation exists pending a review of the supplier's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.2.3 Test Procedures and Test Details. Test procedures in accordance with MIL-STD-781 to be used for each phase of the reliability assurance tests shall be prepared by the supplier. Each procedure shall be submitted to and approved by the procuring activity before that phase of the reliability assurance testing contained in the procedure may begin. The test procedures shall contain all pertinent test details such as length of the duty cycle, length of the cooling and heating portions of the cycle, performance characteristics to be measured, failure criteria, burn-in period, test equipment and wiring diagrams, etc. The following paragraphs shall be considered as minimum requirements and shall apply to all phases.

4.4.2.3.1 Duty Cycle. The duty cycle shall consist of supplying each type indicator with a signal input to each channel that will cause an upscale and downscale excursion of the tapes covering at least two-thirds of the full scale range at a rate of 4 $\pm$ 1 cycles per minute. The 5v lighting circuit shall be energized when the indicator is operating.

4.4.2.3.2 Performance Characteristics to be Measured.

4.4.2.3.2.1 Reliability Qualification Phase Test. The scale error test (4.7.4) shall be conducted at least once each week on indicators being subjected to the Reliability Qualification Phase Test and shall not exceed the tolerances listed in 3.4 for the Reliability Qualification Phase for the type indicator being tested. In addition, once each working day the indicators shall be observed for erratic operation, lamp failure, or catastropic failure. At the end of the test, the indicators shall conform to the requirements of the Individual Tests as amended by this paragraph.

4.4.2.3.2.2 Reliability Production Acceptance Phase Test. Once each working day during the Burn-In Period and the Reliability Test Phase, the indicators shall be observed for erratic operation, lamp failure, or catastropic failure. At the end of the Burn-In, the indicators shall be subjected to the scale error test (4.7.4). At the end of the Reliability Test Phase, the indicators shall conform to all requirements of the Individual Tests.

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4.4.2.3.3 Failure Criteria. Whenever performance characteristics fall below the requirements of 4.4.2.3.2 for the phase testing being conducted, at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure unless the procuring activity and the supplier agree that one part caused the other parts of the same indicator to fail.

4.4.2.3.4 Preventative Maintenance. No preventative maintenance may be accomplished on the indicators while they are on test. The test program may be interrupted to perform preventative maintenance on the test equipment, provided it is recorded in the test log.

4.4.3 Individual Tests. The individual tests are as follows:

Examination of Product
Warmup
Operation
Scale Error
Starting (RPM only)
Position Error
Response Time
Lighting (Individual Tests of MIL-L-25467)
Power Consumption
Sealing
Precipitation or Fog

4.4.4 Sampling Plans. The sampling plans shall consist of Sampling Plans "A" and "B". Indicators selected for the sampling plan tests shall first have passed the Reliability Production Acceptance Phase Test. Indicators which have been subjected to Sampling Plan "A" Tests shall not be delivered on contract until they have been refurbished and have passed the Reliability Production Acceptance Phase Test. Indicators which have been subjected to the Sampling Plan "B" Tests shall not be delivered on contract.

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4.4.4.1 Sampling Plan "A" Selection. Each type indicator being submitted for Sampling Plan "A" Tests shall be selected at random by the Government Quality Assurance Representative in accordance with the following schedule:

Quantity of type indicator offered for acceptance	Quantity to be selected for Plan "A" Tests
First 25	(See note)
26 to 100	1
101 to 200	1
Each additional 100 or fraction thereof greater than 25	1

NOTE: When Sampling Plan "B" is invoked, the quantity shall be zero. When Sampling Plan "B" is waived, the quantity shall be one.

When an indicator failure is detected, no indicators of the set still on hand or later produced shall be accepted until the extent and cause of failure have been determined. If the failure is determined to be inherent to one type of indicator only, the other types of indicator may be accepted. Appropriate corrective action, approved by the procuring activity, shall be taken prior to further acceptance of the types of indicator affected by the failure. Also, when a failure occurs, the sampling procedure for the affected types shall start a new cycle; that is, one of the next 25 produced, etc.

4.4.4.1.1 Sampling Plan "A" Tests. Each indicator selected for Sampling Plan "A" shall be subjected to the following tests:

Individual Tests
Magnetic Effect
Low Temperature Operation
High Temperature Operation
Voltage and Frequency Variation
Lighting (Sampling Plan "A" Tests of MIL-L-25467)
Thermocouple Circuit Resistance Effect (EGT only)
Grounding of Thermocouple Extension Wires (EGT only)
Stray Signal and Beat Frequency Toleration (EGT only)
Power Frequency Susceptibility Test (RPM only)
Voltage Surge

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4.4.4.1.2 Disposition of Sampling Plan "A" Indicators. Upon completion of Sampling Plan "A" Tests, a sample indicator may be delivered on contract provided it meets all of the following requirements:

a. It is been refurbished or is in "good as new" condition (see 6.3.2 and 6.3.3).

b. It is representative of production indicators currently being accepted.

c. It is otherwise satisfactory.

4.4.4.2 Sampling Plan "B" Instructions. One of each type indicator on the contract shall be selected at random from the first 25 conforming to the Reliability Production Acceptance Phase Test and shall be submitted within 10 days after selection by the Government Quality Assurance Representative. These indicators shall be forwarded at the supplier's expense to a Government laboratory designated by the procuring activity.

4.4.4.2.1 Sampling Plan "B" Approval. Approval of Sampling Plan "B" indicators shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material, or performance defect made evident during these tests shall be corrected by the supplier to the satisfaction of the procuring activity. Failure of any type indicator to pass any of the tests may be cause for deliveries of all types, and shall be cause for that type indicator under the contract to cease until proper corrective action, approved by the procuring activity, is accomplished.

4.4.4.2.2 Sampling Plan "B" Tests. Each indicator selected for Sampling Plan "B" Tests shall be subjected to the following tests:

Sampling Plan "A" Tests
Vibration
Low Temperature Exposure
High Temperature Exposure
Electromagnetic Interference
Temperature Shock
Salt Fog
Shock
Test for Helium
Crash Safety
Contrast and Lamp Circuit (MIL-L-25467)
Internal Examination

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4.5 Standard Conditions for Test. Unless otherwise specified, all tests required by this specification shall be made under the following conditions:

Temperature	Room ambient (25 ± 5°C)
Pressure	Normal atmospheric (approximately 29.92 inches of mercury)
Humidity	Room ambient up to 90 percent relative humidity
Instrument vibration	None
Input Power	26 ± 1 v ac, 400 ± 5 Hz
Lighting	5 ± 0.5 v dc or 400 ± 5 Hz ac

4.5.1 Attitude. Unless otherwise specified, the indicators shall be tested in their normal operating positions with the lateral and longitudinal case axes in the horizontal and the scale upright.

4.5.2 Mounting. The indicators shall be mounted in a mounting rack assembly in accordance with Figure 8 for the vibration, shock, and crash safety tests. The rack shall either be filled with actual indicators or simulated indicators used to provide a complete indicator complement during these tests. The indicators need not be mounted in the rack for any other tests for mounting which would be applicable, if proper orientation and adequate support is provided.

4.5.3 Test Signal Inputs. The test signal inputs shall be from the following sources or equivalent which are infinitely variable throughout the scale range.

4.5.3.1 Thermocouple Simulator. Unless otherwise specified, the input signals shall consist of dc millivoltages derived from a potentiometer circuit. The output terminals from the millivoltage source shall be made of copper and shall be connected to MIL-W-5846, Type I, Class E, No. 20 American Wire Gage Chromel and Alumel Thermocouple Lead Wire. Each junction so formed shall be compensated or maintained at the reference junction temperature listed on the nameplate. The alumel and chromel leads shall terminate at a connector which mates with Connector J2 of 3.4.1.5. The pins of the mating connector shall be alumel and chromel. The source, when measured at the pins of the mating connector shall be accurate to witnin ± 20 microvolts dc.

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4.5.3.2 Synchro Transmitter. A synchro transmitter, accurate to within 3 minutes of arc shall be used. When the leads of the synchro transmitter, labeled to provide the voltage output specified in the note in 3.4.3.6.3, are connected to the FF indicator, as shown in Figure 4, the transmitter shall produce the flow rates listed in Table III.

4.5.3.3 Tachometer. The single-phase test signal shall be derived from a tachometer generator conforming to MIL-G-26611. The tachometer generator test stand shall be capable of measuring the generator shaft speed to within ± 2 rpm.

4.5.4 Lighting. The indicator lighting system shall be energized during all tests in which the indicator is energized unless otherwise specified, and shall operate satisfactorily after completion of each test.

4.5.5 RPM Tolerances. When tolerances for the RPM indicator are given in percent, this percent is percent RPM where 100 percent RPM is equal to 4200 rpm, and not percent full scale or percent of the reading. For example, the tolerances listed for the expanded scale at 25°C are ± 0.4 percent RPM, which is 0.004 X 4200 rpm or ± 17 rpm.

4.6 Individual and Sampling Plan Test Procedures. A detailed test procedure shall be prepared for the tests to be conducted prior to starting the tests. The contractor shall be responsible for preparing the procedure for all of the tests and examinations not conducted at a Government laboratory. The procedure shall be submitted to and approved by the procuring activity prior to use. The procedure shall state exact conditions under which measurements are to be made and shall include the detailed criteria used to evaluate all requirements and performance characteristics. The procedure shall contain all necessary data sheets needed to record the conditions and measurements. The test procedure shall provide for collection and recording of quantitative data where quantitative requirements and tolerances are specified. MIL-T-18303 shall be used as a guide.

4.7 Test Requirements.

4.7.1 Examination of Product. Each indicator shall be examined externally to determine conformance with all applicable drawings and with all the requirements of this specification and other applicable specifications not covered by tests.

4.7.2 Warmup. The indicator shall be stabilized at standard conditions for a minimum period of 4 hours without power or signal inputs applied. Standard input power and signal inputs to provide full scale indication shall be instantaneously applied to both channels of the indicator. Within 10 seconds the tapes shall respond, producing a full scale stabilized indication of the signal inputs.

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4.7.3 Operation. The indicator signal inputs shall be adjusted to allow qualitative determination of the maximum tape drive stops in accordance with 3.3.11.

The signal inputs shall then be set to produce full scale tape indications and the signal inputs interrupted instantaneously, first by open-circuiting and then by short-circuiting. Under both conditions of input signal interruption, the tape shall drive to zero indication within 5 seconds, but the power-off warning shall not be displayed.

Signal inputs shall then be restored to produce full scale tape indications and the 26v power inputs interrupted by open-circuiting. The tapes shall drive to an off-scale position and the word "OFF" shall be displayed within 5 seconds.

The 26v power inputs shall be reapplied, one channel at a time and the tapes monitored to determine that the indicator mechanisms are damped to prevent more than one overshoot and are stabilized to prevent hunting or tape oscillation.

4.7.4 Scale Error. The signal inputs shall be adjusted to drive the top of the white tapes to positions of alignment with the centers of the scale markings for the test points listed below, first in the upscale direction and then in the downscale direction. The difference between the signal input values required to produce these indications and the values listed in 3.4 for the type indicator being tested shall not exceed the signal input equivalent to the indicator tolerances listed for 25°C.

TABLE IV. Test Points.

RPM (Percent)	EGT (°C)	FF (pph)
15	0	500
20	200	1000
40	410	2000
71	500	3000
80	600	3200
90	690	4000
100	900	4900
110		9000

4.7.5 Starting (RPM only). The signal inputs to each channel of the RPM indicator shall be slowly increased from zero rpm. The speed at which the tapes provide stable indications shall be no greater than 420 rpm generator shaft speed.

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4.7.6 Position Error. The signal inputs shall be adjusted to position the tapes at any of the test points listed in 4.7.4 that are in the expanded scale range with the indicator in the normal operating position (4.5.1). The indicator shall then be moved slowly from the normal position to the test positions specified below, while the indications are continuously observed. The maximum difference in indications from the normal position indication shall be recorded. The indications at each of the test positions shall also be recorded. The maximum deviation from the normal position shall not exceed the tolerance listed in 3.4 for the type indicator being tested.

a. Rotated from normal operating position through 90 degrees until the dial is face up.

b. Rotated from normal operating position through 90 degrees until the dial is face down.

c. Rotated from normal operating position through 360 degrees with the dial remaining vertical. Readings shall be taken at 90 degree increments and at the point of maximum difference in indication.

4.7.7 Response Time. Input signals shall be applied to the indicator to produce minimum scale indications. The input signals shall then be instantaneously increased to the values required to produce full-scale indication of the tapes. The tapes shall respond smoothly without hesitation from minimum to full-scale indication within 5 seconds.

4.7.8 Lighting (Individual Tests of MIL-L-25467). The indicator shall be subjected to and meet the requirements of the individual tests of MIL-L-25467.

4.7.9 Power Consumption. The indicator shall be energized with approximately midscale signal input applied to both channels for a minimum period of five minutes. At the end of this period the static power consumption shall be measured and shall not exceed 9.5 va total consumption for the 26v circuits and 3 va for the lighting circuits. The indicator tapes shall then be set at minimum scale indication and the signal inputs instantaneously changed to produce full scale tape indication. The power consumption of the 26v circuits shall be measured during tape travel and the dynamic power consumption shall not exceed 14 va. The power factor of the 26v power circuits shall be within 0.85 lagging to 0.95 leading.

4.7.10 Sealing. The indicator shall be tested for case leakage using a mass spectrometer helium leak detector or by an equivalent, at least equally sensitive, quantitative leakage test. The total leakage rate of the indicator shall not exceed 1 micron cubic foot per hour (0.1 micron cubic foot per hour indicated leak rate using a mass spectrometer helium leak detector) at a pressure differential of 1 atmosphere.

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4.7.11 Precipitation or Fog. The indicator shall be placed in a temperature chamber. The temperature of the circulating air shall be 85 ± 2°C and the indicator shall be kept in the chamber for a minimum period of 2 hours. The indicator shall then be removed from the chamber and within 30 seconds be immersed face down in a water bath to a depth of 1/2 ± 1/4 inch for a period of 3 ± 0.1 minutes. The depth of the water bath shall be a minimum of 3 inches and the volume of water shall be such that during the immersion period, with the water being stirred, the temperature of the water does not exceed 5°C. Immediately thereafter, the display outer surface shall be wiped dry (compressed air shall not be used) and the indicator inspected in the normal operating position, for a minimum period of 3 minutes for evidence of water or oil fog. There shall be no evidence of moisture or other deposits on the inner surface of the cover glass.

4.7.12 Magnetic Effect. The indicator, first not operating and then operating at approximately 90 percent of full-scale indication, shall be positioned to place the nearest point of the indicator 5 1/2 inches from a shortbar magnetic compass along the magnetic east or west of the compass. The compensating magnets shall have been removed from the compass. The horizontal field intensity shall be between 0.17 and 0.19 oersted. The indicator shall be positioned such that its longitudinal axis is parallel to the compass pointer and shall be slowly rotated 360 degrees about its longitudinal axis. The compass pointer deflection during indicator positioning and rotation shall at no time exceed 3 degrees.

4.7.13 Low Temperature Operation. The indicator, not energized, shall be subjected to a temperature of -54° ± 2°C for a minimum period of 4 hours. While still maintained at this temperature, power and signal inputs shall be applied and at the end of a 2-minute warmup period, the indicator shall be subjected to the scale error test and shall meet the tolerances listed for -54°C in 3.4 for the type indicator being tested.

4.7.14 High Temperature Operation. The indicator, while energized, and with approximately full-scale signal input applied to both channels, shall be subjected to a temperature of 71° ± 2°C for a minimum period of 4 hours, and then be tested at this temperature for scale errors. The scale errors shall not exceed the tolerances listed for 71°C in 3.4 for the type indicator being tested.

4.7.15 Voltage and Frequency Variation. The indicator shall be subjected to a scale error test while energized at the following combinations of voltage and frequency. The scale errors shall not exceed the tolerances listed for -54°C in 3.4 for the type indicator being tested.

a. 23.4v, + 0.5v, - 0v ac; 380 ± 5 Hz

b. 23.4v, + 0.5v, - 0v ac; 420 ± 5 Hz

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c. 28.6v, + 0v, - 0.5v ac; 380 ± 5 Hz

d. 28.6v, + 0v, - 0.5v ac; 420 ± 5 Hz

4.7.16 Lighting (Sampling Plan "A" Tests of MIL-L-25467). The indicators shall be subjected to and meet the requirements of the Sampling Plan "A"Tests of MIL-L-25467.

4.7.17 Thermocouple Circuit Resistance Effect (EGT only). Each channel of the EGT indicator shall be subjected to a signal lead resistance change from 5 ohms to 200 ohms at the 500°C and 600°C test points. The change in indication shall not exceed 2°C.

4.7.18 Grounding of Thermocouple Extension Wires (EGT only). The Chromel or (+) wire and Alumel or minus (-) wire shall be alternately connected to the grounded or low potential side of the power supply for a period of 1 minute. Prior to grounding the wire, the scale error shall be determined at the 500°C and 600°C test points and the full-scale response time shall be determined. The scale error at these test points and the response time test shall be repeated with each lead alternately grounded. The scale errors with either lead grounded shall not differ from the scale errors without a lead grounded by more than 2°C. The response times with the leads alternately grounded shall not exceed 4 seconds.

4.7.19 Stray Signal and Beat Frequency Toleration (EGT only). The EGT indicator shall be connected as shown on Figure 9. The indicator shall be a minimum distance of 6 feet from the test setup during this test with the bundled cable extended from the rear of the case in a straight line for a minimum of 3 feet. All switches shall be opened. With the 115v ac, 400 Hz power supply turned off, standard power inputs and thermocouple signals to produce indications of 490°C shall be applied to the channels of the indicator. After 2 minutes with the 490°C indications, the cathode ray oscilloscope and the 115v ac power supply shall be turned on and with a frequency of 380 Hz, the frequency of the variable frequency supply shall be increased at a rate not exceeding 1 Hz each 5 seconds until 420 Hz is reached. The indicator tapes shall be closely observed throughout this test. This procedure shall be repeated with only the S-1 switch closed; S-1 and S-2 switches closed; S-1, S-2, and S-3 switches closed; S-1, S-2, S-3, and S-4 switches closed; and then with all switches closed. This test shall be repeated with the thermocouple leads alternately grounded. The grounding may be accomplished at any point along the thermocouple leads between the mating connector and the signal source, and any common ground may be used. Tape position changes and tape oscillation shall be noted. The maximum change in tape position shall not exceed 2°C. Tape oscillation shall be cause for rejection of the indicator.

4.7.20 Power Frequency Susceptibility Test (RPM only). The RPM indicator shall have a 400 Hz signal of 2v rms applied across the signal inputs.

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when either the Chromel or Alumel conductor is grounded separately at any

With the 2v modulation signal applied, the scale error test shall be conducted. Scale errors shall not exceed the tolerances listed for 25°C in 3.4.2.6.4 and there shall be no tape oscillation.

4.7.21 Voltage Surge. The tapes of the indicators shall be set at approximately midscale and the 26v power inputs adjusted to 26 ± 1v, 420 ± 2 Hz. A voltage surge shall be applied to these power inputs to provide momentary voltage of 40 ± 1v and return to 26 ± 1v. The duration of the surge, including rise and decay time, shall be 0.090 ± 0.005 second. The frequency shall then be decreased to 400 ± 5 Hz and a voltage surge (overvoltage) of 36 ± 1v applied to the power inputs for 0.90 ± 0.05 seconds. The two transient voltage applications shall be spaced at least one minute apart. The indicator shall then be subjected to and comply with the scale error test of 4.7.4.

4.7.22 Vibration. The indicators, installed in a mounting rack assembly, shall be subjected to a vibration resonance search, resonance dwell, and cycling test in accordance with Procedure I, Part 1 of Method 514.2, MIL-STD-810 for schedule b.1 equipment. The test procedure and test schedule listed for equipment without vibration isolators of Table 514.2-II and Curve C of Figure 514.2-2 shall be used. Prior to conducting the resonance test in each axis, the indicator tapes shall be positioned at the indications specified below. The tapes shall be monitored during the resonance search and dwell, and the change in indication plus oscillation shall not exceed the tolerance specified in 3.4 for that type indicator.

The vibration cycling in a particular axis may be conducted concurrently with the resonance test in that axis. Upon completion of vibration, the indicators shall be subjected to and meet the scale error requirements of 4.7.4. No looseness in the mechanism or parts nor damage to the indicators shall result from this test.

a. FF 3000 pph

b. RPM 100% RPM

c. EGT 450°C

4.7.23 Low Temperature Exposure. The indicator shall be tested in accordance with Procedure I, Method 502.1 of MIL-STD-810. Following exposure at -65°C for 24 hours, the temperature shall be adjusted to -54°C and maintained for a minimum of 4 hours. Power and signal inputs shall be applied and following a 2-minute warmup, the indicator shall be subjected to the scale error test. The tolerances listed in 3.4 for -54°C for the type indicator being tested shall apply. The indicators shall then be returned to room temperature and allowed to stabilize for a period of not less than 4 hours and then again be subjected to the scale error test. The tolerances listed in 3.4 for 25°C for the type indicator being tested shall apply.

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The indicator shall be examined for deterioration or damage due to extreme temperature exposure. No damage or deterioration shall have occurred which will affect subsequent operation.

4.7.24 High Temperature Exposure. The indicator shall be tested in accordance with Procedure I of Method 501.1 of MIL-STD-810. Following exposure at 95°C for 48 hours, the temperature shall be adjusted to 71°C and maintained for a minimum of 4 hours. Power and signal inputs shall be applied, and following a 1-minute period for equipment warmup, the indicator shall be subjected to the scale error test. The tolerances listed in 3.4 for 71°C for the type indicator being tested shall apply. The indicator shall then be maintained at room temperature for a period of not less than 4 hours and shall then again be subjected to the scale error test. The tolerances listed in 3.4 for 25°C for the type indicator being tested shall apply. The indicator shall be examined externally for deterioration or damage due to extreme temperature exposure. No damage or deterioration shall have occurred which will affect subsequent operation.

4.7.25 Electromagnetic Interference. The indicator shall be subjected to, and meet, the electromagnetic interference tests in accordance with MIL-STD-462 (see 3.3.15).

4.7.26 Temperature Shock. The indicator shall be tested in accordance with Method 503.1 of MIL-STD-810. Following stabilization at room temperature for a period of not less than 4 hours at the conclusion of the test, the indicator shall be subjected to the scale error test. The tolerances listed in 3.4 for 25°C for the type indicator being tested shall apply. The indicator shall be examined externally for deterioration or damage due to this test. No damage or deterioration which will affect subsequent operation shall have occurred.

4.7.27 Salt Fog. The indicator, with external connections made in such a manner as to simulate installed conditions, shall be subjected to a 48-hour salt fog test in accordance with Procedure I, Method 509.1 of MIL-STD-810. Immediately after removal from the salt fog chamber, the indicator shall be washed with distilled water and air dried. Within 2 hours after removal from the salt fog chamber and again at the end of 50 hours, the indicator shall be subjected to the scale error test. The tolerances listed in 3.4 for 25°C for the type indicator being tested shall apply. The indicator shall be examined and there shall be no evidence of external corrosion or deterioration which will affect subsequent operation.

4.7.28 Shock. The indicators, installed in a mounting rack assembly and operating at the indications listed in 4.7.22, shall be subjected to a shock test in accordance with Procedure I for Flight Vehicle Equipment of Figure 516.2-2 of MIL-STD-810. Following this test, the tests specified in 4.7.3, 4.7.4, and 4.7.10 shall be conducted. The indicators shall

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conform to the requirements therein. The indicators shall be examined and there shall be no deformation of the case or damage to the indicator as a result of this test.

4.7.29 Test for Helium. The filling tube of the indicator shall be cut and the indicator tested for the presence of helium with a mass spectrometer helium detector. Failure to detect helium shall be cause for rejection.

4.7.30 Crash Safety. The indicators, installed in a mounting rack assembly, shall be subjected to a crash safety test, Procedure III, Method 516.2 of MIL-STD-810 for Flight Vehicle Equipment of Figure 516.2-2. Bending and distortion of the indicator case is permitted, however, failure at the mounting points, case rupture, or part separation which is not retained within the case enclosure shall be cause for rejection.

4.7.31 Contrast and Lamp Circuit. The indicator shall be subjected to, and meet, all of the contrast and lamp circuit requirements specified in MIL-L-25467.

4.7.32 Internal Examination. The indicator case shall be removed and the indicator mechanism examined for conformance to the requirements of Section 3. There shall be no evidence of corrosion or deterioration other than that caused by the crash safety test.

4.8 Packaging, Packing, and Marking. As indicators are being prepared for deliveries, an inspection shall be made to determine that the packaging, packing, and marking conforms to the requirements of Section 5. Defective marking, packaging, materials, or packing shall not be accepted.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The indicator shall be packaged in accordance with MIL-STD-794, Level A or C, as specified in the contract or order. For Level A packaging, the method of preservation shall be in accordance with MIL-P-116, Method 1A-5, without preservation compound, using metal reusable containers.

5.2 Packing. The indicators shall be packed in accordance with MIL-STD-794, Level A, B, or C, as specified in the contract or order.

5.3 Marking. The interior and exterior containers shall be marked as specified in MIL-STD-794 with the additional precautionary marking:

FRAGILE
DELICATE INSTRUMENT
HANDLE WITH CARE

appearing conspicuously on the two opposite sides of each interior package and exterior shipping container.

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5.3.1 Re-inspection Date. The re-inspection date markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended Use. The indicator set covered by this specification is intended for use in A6A or A6E aircraft powered by two Pratt & Whitney J52-P-6 or J52-P-8 turbojet engines.

6.2 Ordering Data. Procurement documents should specify all of the procurement, contract data, and procedure requirements necessary to assure receipt of a desirable end product.

6.2.1 Procurement Requirements.

a. Title, number, and date of this specification.

b. Quantity and type designation of indicators desired (see 1.2).

c. Levels of packaging and packing (see 5).

d. The re-inspection date to be marked on interior packages and exterior shipping containers (see 5.3.1).

e. Laboratory that will conduct First Article and Sampling Plan "B" Tests to be conducted at a Government laboratory (see 4.3 and 4.4).

6.2.2 Contract Data and Procedure Requirements. The data and procedure requirements, summarized below, should be listed on a Form DD 1423 and incorporated into the contract.

a. First Article Individual Test Data (4.3).

b. First Article Reliability Qualification Phase Test Report (4.3.1).

c. Contractor conducted Quality Conformance Test Reports (Sampling Plan "A" Tests) (4.4).

d. First Article Reliability Qualification Phase Test Procedure and Reliability Production Acceptance Phase Test Procedure (4.4.2.3).

e. Individual Test Procedure and Sampling Plan "A" Test Procedure (4.6).

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6.3 Definitions. The following definitions supplement those in MIL-STD-109 for terms used herein.

6.3.1 Double Amplitude. The term "double amplitude" as used in this specification means total displacement as measured from one extreme to the opposite extreme.

6.3.2 Refurbished. "Refurbished" shall mean completely overhauling the indicator such that all component parts meet the current new parts standards, and shall refer only to indicators which, after being overhauled, have been subjected to, and met, all the requirements of a new instrument.

6.3.3 Good as New. A "good as new" condition shall mean indicators operated less than 5 percent of the specified MTBF (350 hours).

6.3.4 Hermetic Seal. A hermetic seal shall be an airtight seal made between metallic and metallic or between metallic and vitric materials. A seal dependent upon gaskets shall not be considered as a hermetic seal.

6.3.5 Corrosion Resistant. "Corrosion resistant" and "protected against corrosion" as used in this specification shall mean that a material exhibits no unaided visual evidence of corrosion subsequent to exposure to a salt fog test of severity and duration at least equal to the test specified in 4.7.27. When tests are performed to determine corrosion resistance during materials selection, the material shall be cleaned by ultrasonic methods or equivalent, just prior to test. Completed indicators shall be tested for corrosion resistance in the "as is" condition.

6.3.6 Indicator. A device which includes all the electronic circuitry, mechanical assemblies, and display required to directly receive and process the output signals from two sensors and produce a visual presentation of the parameter being measured by each sensor.

6.3.7 Indicator Channel. That portion of an indicator that includes all the electronic circuitry, mechanical assemblies, and display required to receive and process the output signal from one sensor and to produce a visual presentation of the parameter being monitored. Usually used to refer to one channel of a multichannel or multisensor indicator. Also channel.

6.4 Precedence of Documents. When the requirements of the contract, this specification, or other applicable subsidiary specifications are in conflict, the following precedence applies:

a. Contract. The contract shall take precedence over any specifiation.

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b. This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications, where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.5 Basic Design Engineer. The basic design engineer for this equipment is Commander, Naval Air Systems Command (AIR-53351), Department of the Navy, Washington, D.C. 20361.

User activity:
Navy - AS

Preparing activity:
Navy - AS

Project No. 6695-NO53

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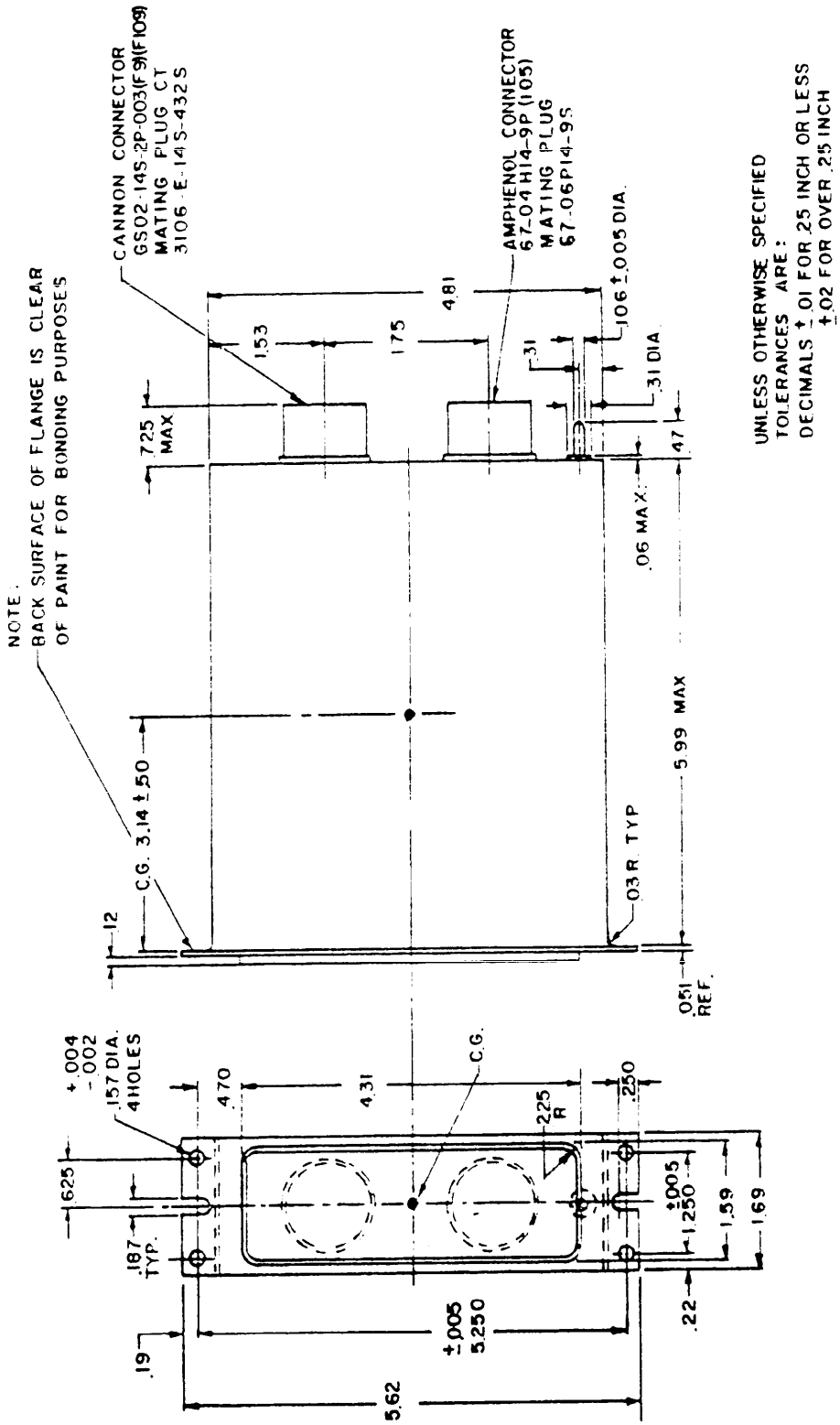


FIGURE 1. EGT indicator case dimensions.

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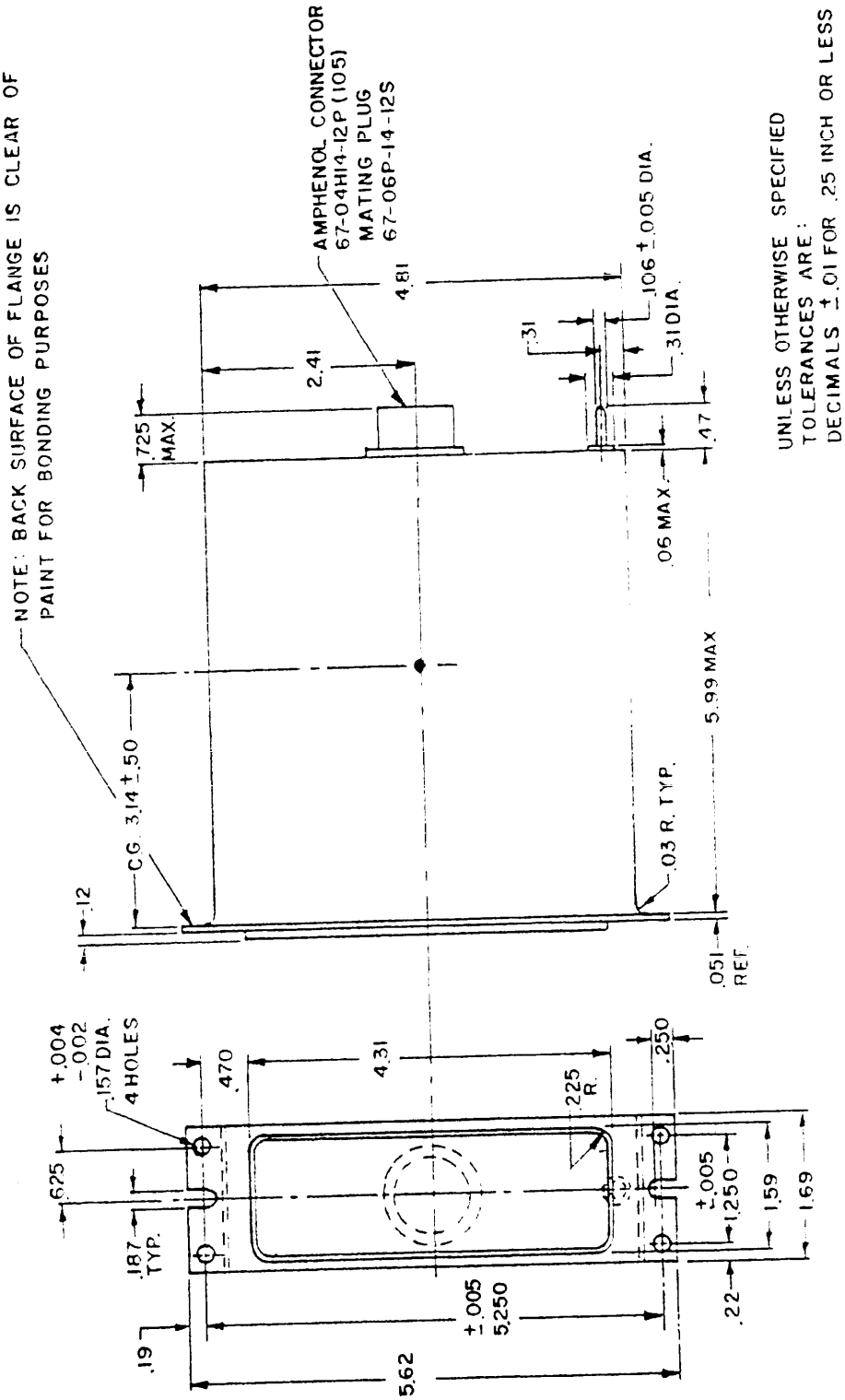


FIGURE 2. RPM indicator case dimensions.

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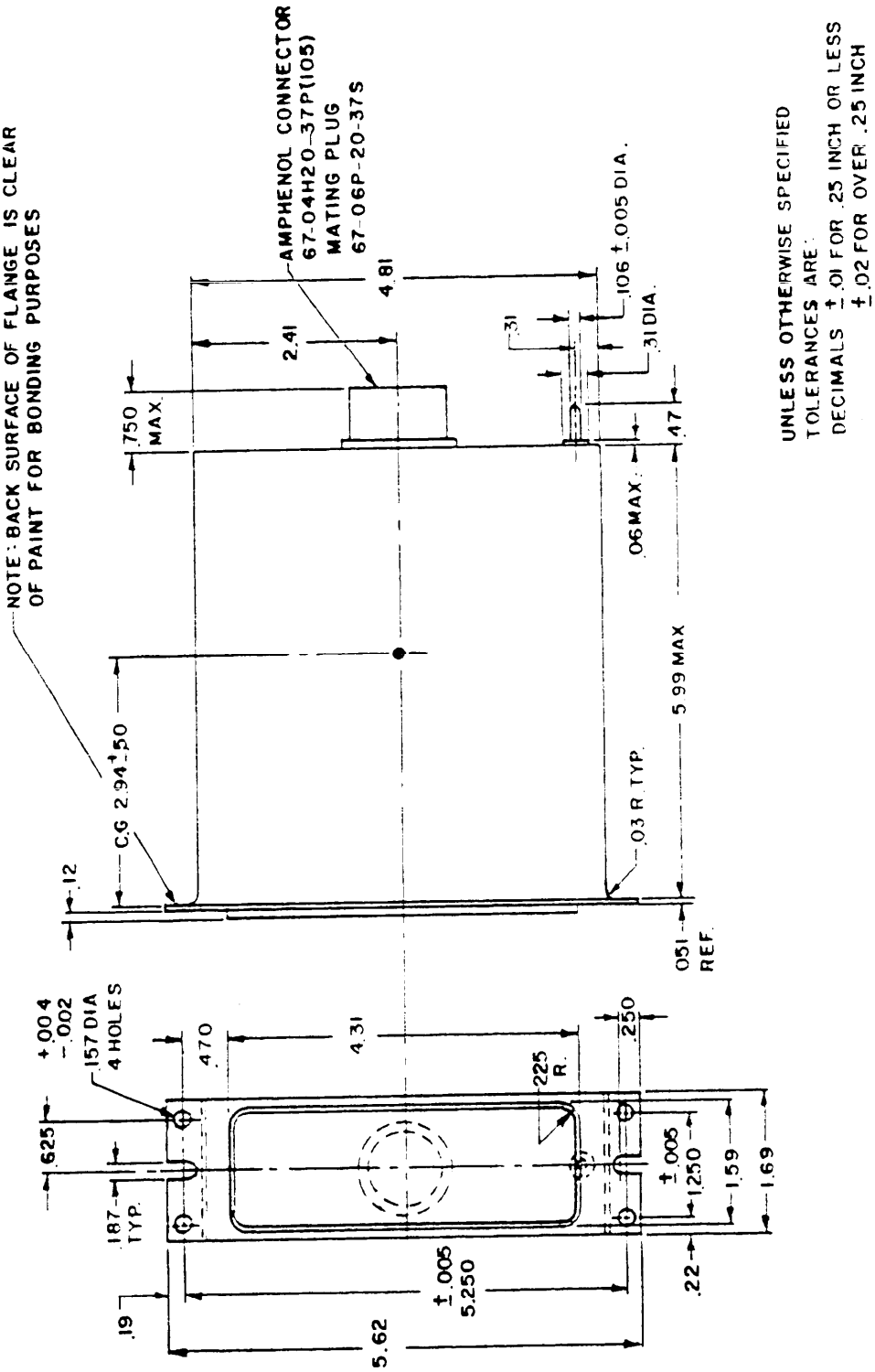


FIGURE 3. FF indicator case dimensions.

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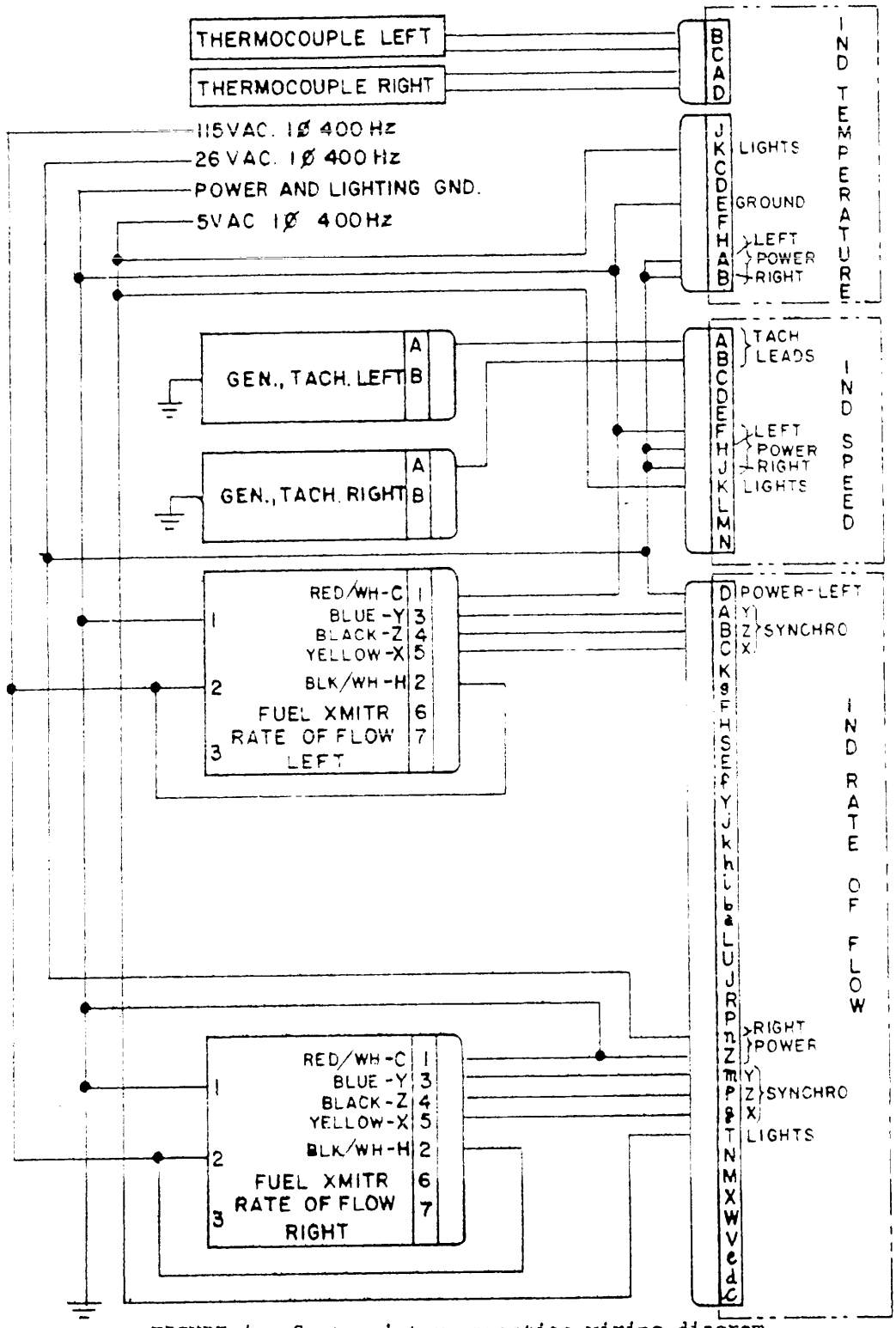
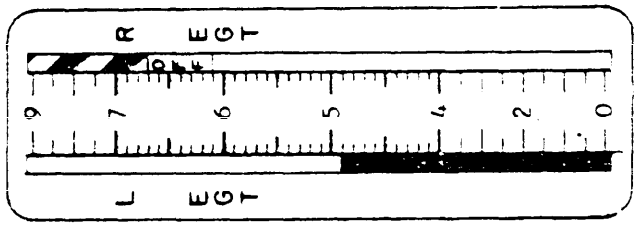


FIGURE 4. System interconnection wiring diagram.

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CAL. MARKS	DIM. (INCHES)
0	.015
-	.153
-	.296
-	.435
2	.572
-	.712
-	.852
-	.995
4	1.139
-	1.211
-	1.283
-	1.357
-	1.429
-	1.501
-	1.575
-	1.647
-	1.720
-	1.794
5	1.868
-	1.940
-	2.011
-	2.085
-	2.159
-	2.233
-	2.305
-	2.378
-	2.452
-	2.526

CAL. MARKS	DIM. (INCHES)
6	2.598
-	2.671
-	2.743
-	2.817
-	2.889
-	2.961
-	3.033
-	3.105
-	3.177
-	3.250
7	3.322
-	3.466
-	3.608
-	3.748
-	3.886
9	

NOTES:

1. ALL LETTERS AND NUMERALS PER MS 33558.
2. BACKGROUND - DULL BLACK
3. MARKING - WHITE

CAL. MARKS AT 400,500,600,700: .030 WIDE X 5/32 LONG

CAL. MARKS AT 0,100,200,300,800,900: .030 WIDE X 5/32 LONG

CAL. MARKS AT 50,150,250,350,750,850: .020 WIDE X 3/32 LONG

CAL. MARKS AT 450,550,650: .030 WIDE X 7/64 LONG

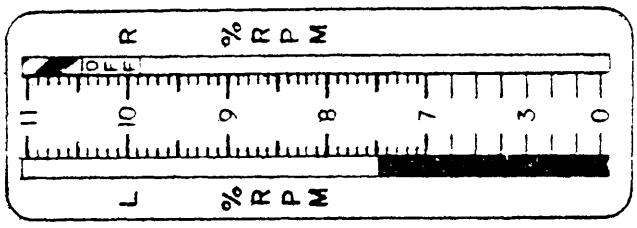
MINOR CAL. MARKS FROM 400 TO 700: .020 WIDE X 1/16 LONG

NUMERALS, L,R,EGT: 1/8 HIGH

"OFF": 3/32 HIGH

FIGURE 5. EGT indicator display.

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CAL. MARKS	DIM. INCHES
0	.015
-	.184
-	.353
3	.521
-	.690
-	.858
-	1.026
7	1.194
-	1.261
-	1.329
-	1.396
-	1.464
-	1.531
-	1.598
-	1.666
-	1.733
-	1.801
8	1.868
-	1.935
-	2.003
-	2.070
-	2.138
-	2.205
-	2.272
-	2.339
-	2.407
-	2.475
9	2.542
-	2.609
-	2.676
-	2.744
-	2.811

CAL. MARKS	DIM. INCHES
-	2.878
-	2.945
-	3.012
-	3.080
-	3.147
10	3.214
-	3.281
-	3.348
-	3.416
-	3.483
-	3.550
-	3.617
-	3.684
-	3.752
-	3.819
11	3.886

NOTES:

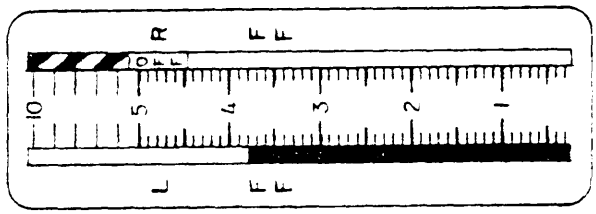
1. ALL LETTERS AND NUMERALS PER MS 33558.
2. BACKGROUND - DULL BLACK
3. MARKING - WHITE

CAL. MARKS AT 7,8,9,10,11: .030 WIDE X 5/32 LONG
CAL. MARKS AT 0,1,2,3,4,5,6: .020 WIDE X 5/32 LONG
CAL. MARKS AT 7.5,8.5,9.5,10.5: .030 WIDE X 7/64 LONG
MINOR CAL. MARKS FROM 7 TO 11: .020 WIDE X 1/16 LONG
NUMERALS, L, R, %, RPM: 1/8 HIGH
"OFF": 3/32 HIGH

FIGURE 6. RPM indicator display.

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CAL. MARKS	DIM. (INCHES)
-	.081
-	.097
-	.162
-	.228
-	.292
-	.359
-	.424
1	.490
-	.556
-	.620
-	.687
-	.751
-	.817
-	.881
-	.948
-	1.014
-	1.079
2	1.145
-	1.211
-	1.276
-	1.342
-	1.407
-	1.473
-	1.538
-	1.604
-	1.669
-	1.735
3	1.800

CAL. MARKS	DIM. (INCHES)
-	1.866
-	1.932
-	1.997
-	2.062
-	2.128
-	2.194
-	2.259
-	2.325
-	2.390
4	2.455
-	2.522
-	2.586
-	2.653
-	2.718
-	2.783
-	2.850
-	2.914
-	2.980
-	3.045
5	3.111
-	3.264
-	3.415
-	3.567
-	3.718
10	3.870

NOTES:

1. ALL LETTERS AND NUMERALS PER MS 33558.
2. BACKGROUND - DULL BLACK
3. MARKING - WHITE

CAL. MARKS AT 1,2,3,4,5: .030 WIDE X 5/32 LONG

CAL. MARKS AT 6,7,8,9,10: .020 WIDE X 5/32 LONG

CAL. MARKS AT .5,1.5,2.5,3.5,4.5: .030 WIDE X 7/64 LONG

MINOR CAL. MARKS FROM 300 to 5000: .020 WIDE X 1/16 LONG

NUMERALS, L,R,FF: 1/8 HIGH

"OFF": 3/32 HIGH

FIGURE 7. FF indicator display.

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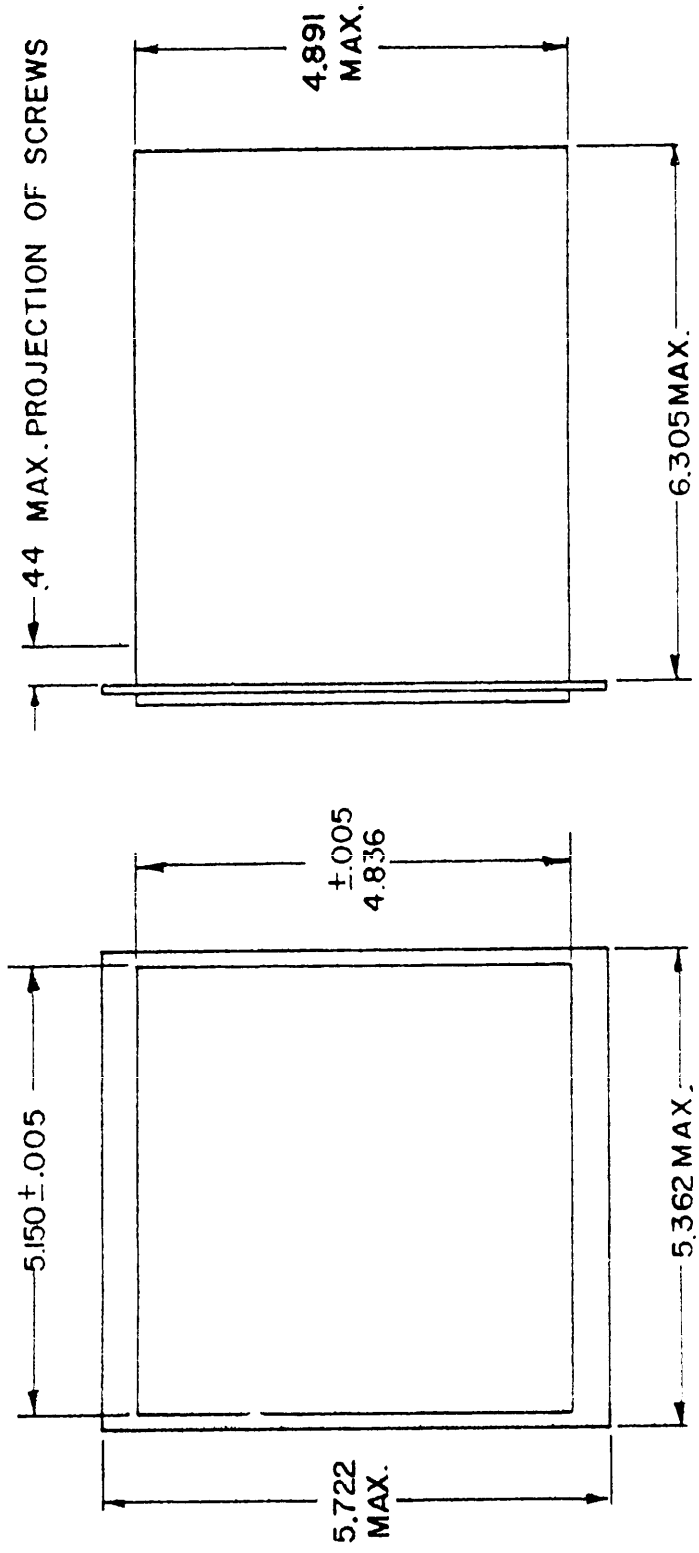


FIGURE 8. Type 128SCAV277 mounting rack outline dimensions.

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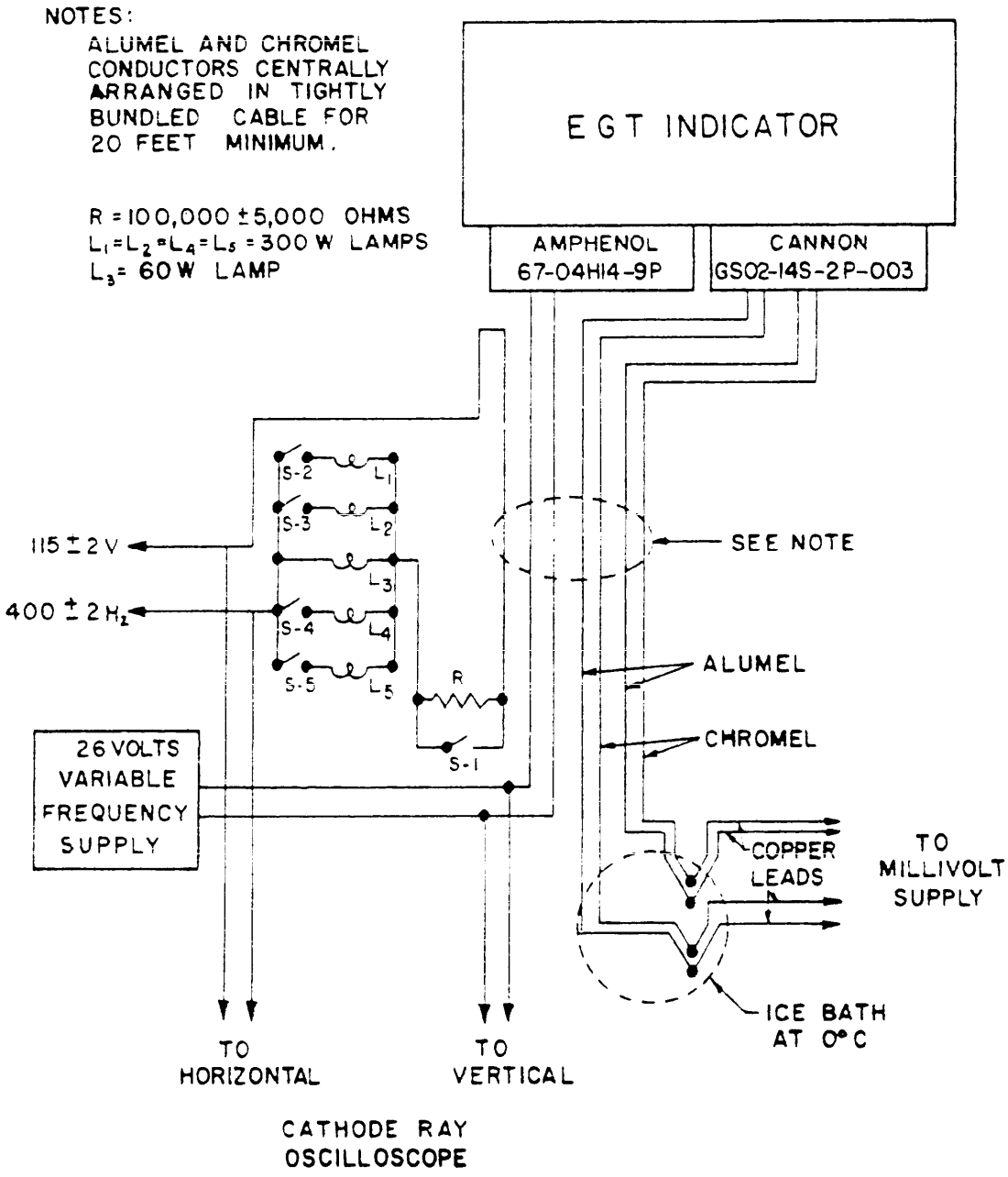


FIGURE 9. Stray signal and beat frequency toleration test setup.

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TABLE V. Metric Equivalents.

Inches	mm	Inches	mm	Inches	mm	Inches	mm
0.002	0.05	0.858	21.79	1.997	50.72	3.012	76.50
0.004	0.10	0.883	22.43	2.003	50.88	3.033	77.04
0.005	0.13	0.948	24.08	2.013	51.13	3.045	77.34
0.01	0.25	0.995	25.27	2.062	52.37	3.080	78.23
0.015	0.38	1.014	25.76	2.070	52.58	3.105	78.87
0.020	0.51	1.026	26.06	2.085	52.96	3.111	79.02
0.030	0.76	1.079	27.41	2.128	54.05	3.14	79.76
0.031	0.79	1.139	28.93	2.138	54.31	3.147	79.93
0.051	1.30	1.145	29.08	2.159	54.84	3.177	80.70
0.06	1.52	1.194	30.33	2.194	55.73	3.214	81.64
0.097	2.46	1.211	30.76	2.205	56.01	3.250	82.55
0.106	2.69	1.250	31.75	2.233	56.72	3.264	82.91
0.12	3.05	1.261	32.03	2.259	57.38	3.281	83.34
0.153	3.89	1.276	32.41	2.272	57.71	3.322	84.38
0.157	3.99	1.283	32.59	2.305	58.55	3.348	85.04
0.162	4.11	1.329	33.76	2.325	59.06	3.415	86.74
0.184	4.67	1.342	34.09	2.339	59.41	3.416	86.77
0.187	4.75	1.357	34.47	2.378	60.40	3.466	88.04
0.19	4.83	1.396	35.46	2.390	60.71	3.483	88.47
0.22	5.59	1.407	35.74	2.407	61.14	3.550	90.17
0.225	5.72	1.429	36.30	2.41	61.21	3.567	90.60
0.228	5.79	1.464	37.19	2.452	62.28	3.608	91.64
0.250	6.35	1.473	37.41	2.455	62.36	3.617	91.87
0.292	7.42	1.501	38.13	2.475	62.87	3.684	93.57
0.296	7.52	1.53	38.86	2.522	64.06	3.718	94.44
0.31	7.87	1.531	38.89	2.526	64.16	3.748	95.20
0.353	8.97	1.538	39.07	2.542	64.57	3.752	95.30
0.359	9.12	1.575	40.01	2.586	65.68	3.819	97.00
0.424	10.77	1.59	40.39	2.598	65.99	3.870	98.30
0.435	11.05	1.598	40.59	2.609	66.27	3.886	98.70
0.44	11.18	1.604	40.74	2.653	67.39	4.31	109.47
0.470	11.94	1.647	41.83	2.671	67.84	4.81	122.17
0.490	12.45	1.666	42.32	2.676	67.97	4.836	122.83
0.50	12.70	1.669	42.39	2.718	69.04	4.891	124.23
0.521	13.23	1.69	42.93	2.743	69.67	5.150	130.81
0.556	14.12	1.720	43.69	2.744	69.70	5.250	133.35
0.572	14.53	1.733	44.02	2.783	70.69	5.362	136.19
0.620	15.75	1.735	44.07	2.811	71.40	5.62	142.75
0.625	15.88	1.75	44.45	2.817	71.55	5.722	145.34
0.687	17.45	1.794	45.57	2.850	72.39	5.99	152.15
0.690	17.53	1.800	45.72	2.878	73.10	6.305	160.15
0.712	18.08	1.801	45.75	2.889	73.38		
0.725	18.42	1.866	47.40	2.914	74.02	1/16	1.59
0.750	19.05	1.868	47.45	2.94	74.68	3/32	2.38
0.751	19.08	1.932	49.07	2.945	74.80	7/64	2.78
0.817	20.75	1.935	49.15	2.961	75.21	1/8	3.18
0.852	21.64	1.940	49.28	2.980	75.69	5/32	3.97

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